GV-200 Series Gate Valve

Model Series: GV-200 | Equipment Type: Cast Steel Gate Valve

1. Overview

The GV-200 Series is a cast steel, rising-stem gate valve intended for isolation service on slurry, water, and process lines in mining and industrial plants. It is rated for full-bore, bi-directional shutoff and is suited to applications with abrasive or particulate-laden media when paired with the optional hardened wedge (GV-200-HW).

2. Technical Specifications

Parameter	Value
Size range	DN50 – DN600 (2" – 24")
Pressure class	ANSI 150 / 300
Body material	Cast steel A216 WCB
Wedge type	Solid flexible wedge
Stem	17-4PH stainless, rising, non-rotating
Seat material	13Cr hardfaced (standard), Stellite (HW option)
Packing	PTFE-impregnated braided graphite, 4-ring set
Max operating temperature	230°C
Actuation	Handwheel (standard), gear operator or electric actuator (optional)

3. Installation Guidelines

1. Confirm line is depressurized, drained, and isolated per site lockout/tagout procedure before removing shipping covers.
2. Inspect flange faces and gasket seating surfaces for damage; clean with a non-abrasive solvent.
3. Install valve with stem vertical ±15° unless the nameplate specifies horizontal mounting.
4. Torque flange bolts in a star (criss-cross) pattern in three incremental passes — see Section 4 for values.
5. Cycle the valve fully open and closed twice before placing into service and verify smooth handwheel travel with no binding.
6. Record installation date, orientation, and line tag on the maintenance log for future reference.

4. Torque Specifications

Apply torque values below using a calibrated torque wrench. Over-torquing bolted connections can distort sealing faces and cause premature gasket or packing failure.

Fastener / Location	Size	Torque (N·m)
Flange bolts, DN50–DN150	M16	130
Flange bolts, DN200–DN300	M20	210
Flange bolts, DN350–DN600	M24	340
Gland follower nuts (packing)	M12	25
Bonnet-to-body bolting	M20	195
Yoke nut (handwheel retaining)	M30	80

5. Maintenance Schedule

Interval	Task
Weekly	Visual inspection for external leakage at stem packing and bonnet gasket.
Monthly	Full-stroke exercise cycle on valves in infrequent-use service to prevent seat sticking.
Quarterly	Check gland packing tightness; re-torque gland nuts to spec if weeping is observed (do not over-tighter
Annually	Remove bonnet, inspect wedge and seat faces for wear/erosion; replace packing set regardless of con
Every 3 years or on failure	Full teardown, NDT inspection of stem, re-lap or replace seats as required.

6. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
Handwheel very stiff to turn	Packing over-tightened, or stem thread dry/corroded	Loosen gland nuts to spec torque; clean and lubricate stem
Valve will not seal fully (passing)	Seat/wedge wear or erosion from abrasive slurry	Inspect wedge and seat faces; refurbish or replace hardfaci
Leakage at stem packing	Packing worn, glazed, or incorrect ring material	Replace full packing set (4 rings) per Section 5; do not add
Valve stuck in open or closed position	Debris lodged in body cavity, or stem bent	Isolate line, remove bonnet, clear debris, inspect stem strai
Audible vibration/chatter when partially open	Valve operated in a throttling position it is not rated for	Gate valves are for isolation devices only — do not use for thro

7. Safety Warnings

■ Never attempt to repack or service a valve under pressure. Fully isolate and depressurize the line first.

■ Gate valves are NOT designed for throttling service; partial-open operation causes accelerated seat erosion and can lead to sudden, uncontrolled failure.

■ Use only OEM-approved packing material rated for the process fluid and temperature.

■ When using a gear operator or electric actuator, verify torque limit switches are set correctly to avoid over-torquing the stem on seating.